

GLOBALLY VALID RELATIONS CONVERTING M_S , m_b AND M_{JMA} TO M_W

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Several magnitude scales are used in earthquake catalogues, compiled by seismological centers all over the world, to give a measure of the “size” of the earthquakes. However, these scales do not behave uniformly for all magnitude ranges. Another defect is that the M_S and m_b scales exhibit saturation effects at different levels for large earthquakes. Both these limitations could result in under- or over-estimation of earthquake magnitudes. Kanamori (1977) and Hanks and Kanamori (1979) proposed the *moment magnitude scale*, M_W , directly connecting the seismic moment with the earthquake magnitude, resulting in a uniform behavior for all magnitude ranges. This definition turns M_W into the most reliable magnitude accurately describing the size of earthquakes.

Many studies have been carried out in different regions and seismotectonic environments to establish relations connecting other magnitude scales to M_W (Heaton et al., 1986; Shedlock, 1999; Papazachos et al., 2002; among many others). The basic objective was to compile earthquake catalogs with all magnitudes expressed in one common scale (M_W) to solve important practical problems (i.e., seismic hazard assessment), as well as theoretical ones (calculation of crustal deformation, etc.).

The major target of the present work is to derive globally valid empirical relations converting magnitudes, expressed in widely used magnitude scales, to equivalent moment magnitudes. Such relations could become a very useful tool in compiling homogeneous earthquake catalogs which are necessary for solving important practical problems (i.e., seismic hazard assessment), as well as theoretical ones (calculation of crustal deformation, etc.). The moment magnitude estimated by Harvard (CMT solutions – 20,775 events from 1976 up to the end of 2003) is used as a reference magnitude. In addition, moment magnitudes for 192 additional events were taken from USGS. Therefore, the earthquakes for which M_W estimations were found reached the total number of 20,967, with magnitudes ranging between 3.1 and 8.7. Their spatial distribution is shown in the map of figure (1).

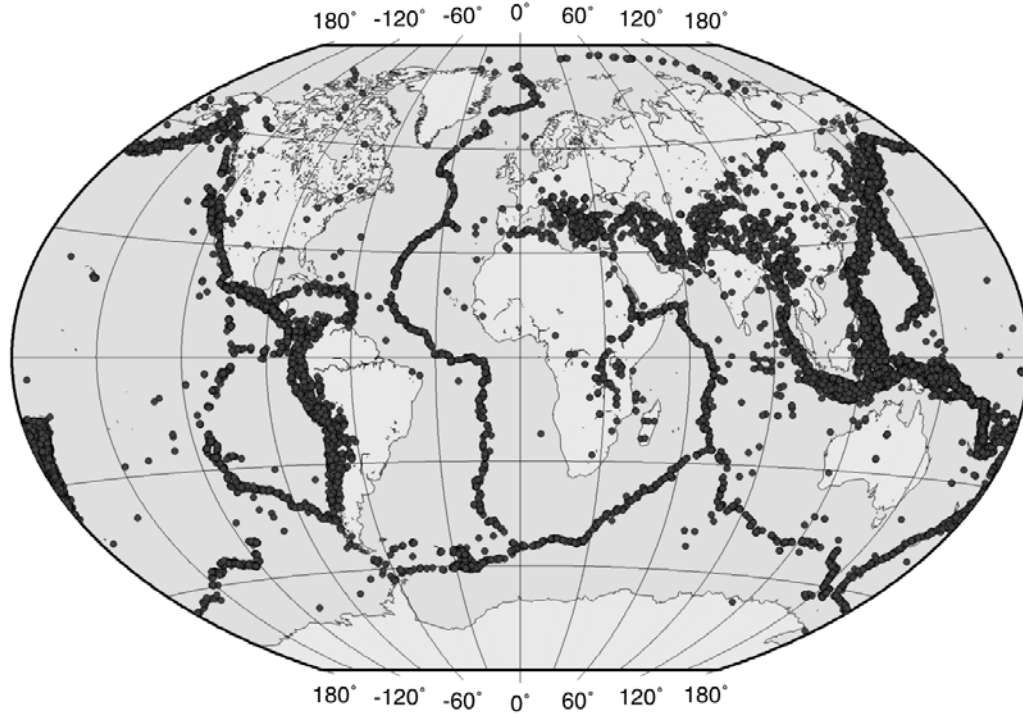


Figure 1.- Spatial distribution of 20,967 earthquakes globally occurred, for which M_W values in the range 3.1-8.7 are available (Scordilis, 2005).

Linear correlation is observed between M_W and M_S (estimated by ISC and NEIC applying the Prague formula, Vanek et al., 1962), clearly divided into two branches, expressed by the following equations (Scordilis, 2005):

$$M_W = 0.65M_S + 2.20, \quad 3.0 \leq M_S \leq 6.1, \quad n = 22415 \quad \sigma = 0.17 \quad (1)$$

$$M_W = 1.00M_S - 0.02, \quad 6.2 \leq M_S \leq 8.0, \quad n = 2340 \quad \sigma = 0.21 \quad (2)$$

The magnitude M_{SK} reported in Karnik (1996) estimated for earthquakes covering the broader area of Europe, is, according to the author, equivalent to M_S . Its comparison with M_W shows clear linear correlation divided into three branches (Scordilis, 2005):

$$M_W = 0.78M_{SK} + 1.45, \quad 4.0 \leq M_{SK} \leq 5.3, \quad \sigma = 0.42 \quad (3)$$

$$M_W = 0.68M_{SK} + 1.99, \quad 5.4 \leq M_{SK} \leq 6.1, \quad \sigma = 0.32 \quad (4)$$

$$M_W = 1.04M_{SK} - 0.30, \quad M_{SK} \geq 6.2, \quad \sigma = 0.34 \quad (5)$$

The relation between m_b (Gutenberg, 1945a, b; Gutenberg and Richter, 1956), estimated by ISC and NEIC, with M_W reveals linear dependency expressed by the relation (Scordilis, 2005):

$$M_W = 0.85m_b + 1.02, \quad 3.5 \leq m_b \leq 6.2, \quad n = 38531, \quad \sigma = 0.29 \quad (6)$$

The m_b magnitude scale exhibits an unstable behavior that could be considered as saturation for earthquakes with $m_b > 6.3$ (or its equivalent $M_W > 6.4$).

All the above relations have been used to estimate M_W^* magnitudes for a big number of earthquakes for which magnitudes from Japan Meteorological Agency (JMA) were given and no original M_W values were available. The calibration of M_{JMA} against original M_W for strong ($M_{JMA} \geq 5.6$) or against M_W^* for weaker ($M_{JMA} \leq 5.5$) earthquakes, revealed two linear relations:

$$M_W = 0.58M_{JMA} + 2.25, \quad 3.0 \leq M_{JMA} \leq 5.5, \quad n = 25305, \quad \sigma = 0.28 \quad (7)$$

$$M_W = 0.97M_{JMA} + 0.04, \quad 5.6 \leq M_{JMA} \leq 8.2, \quad n = 999, \quad \sigma = 0.22 \quad (8)$$

Relation (8) shows that M_{JMA} is almost equivalent to M_W for strong ($M_{JMA} \geq 5.6$) earthquakes.

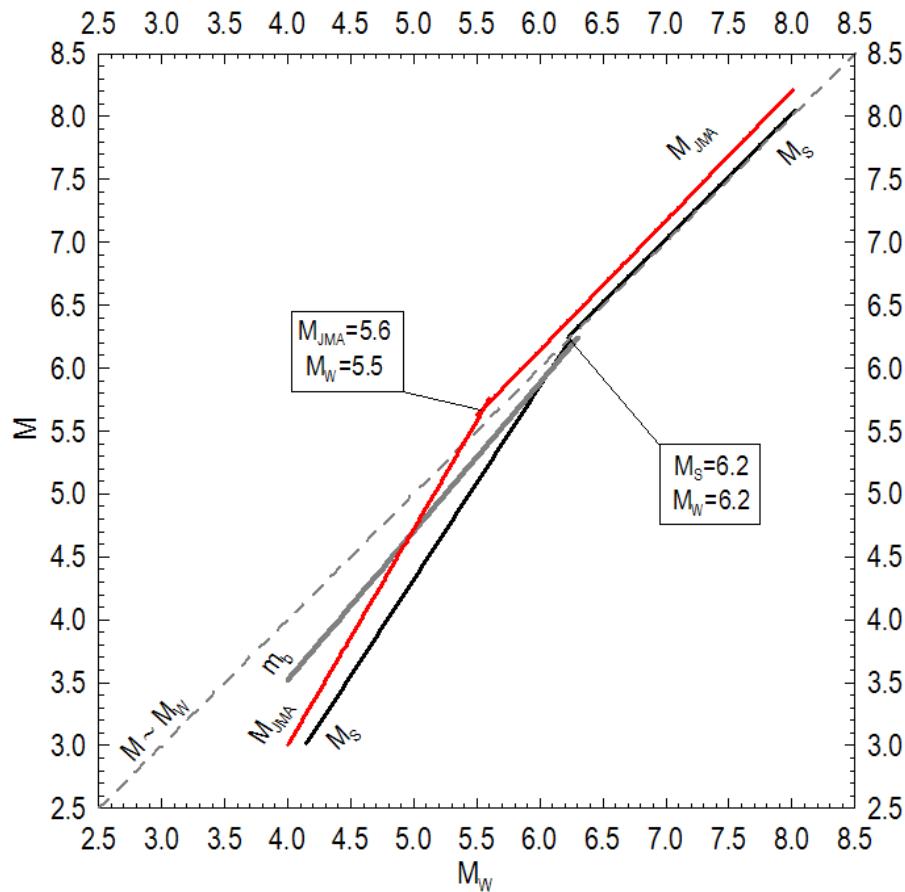


Figure 2.- Summary plot of the final results for the comparison between m_b , M_S and M_{JMA} magnitude scales with M_W .

Different estimates of M_L are given by several seismological centers. The main reasons of this inconsistency are: a) the operation of Wood Anderson seismographs with different effective magnification, smaller than the nominal one ($\sim 2,800$), b) the fact that magnitudes

not estimated from typical Wood-Anderson instruments are almost always reported as M_L and c) the distance corrections applied are often adopted from the original M_L definition and not estimated for each region. For these reasons, no general globally valid relation between M_L and M_W magnitudes can be proposed and such relations must be derived for each group of Wood-Anderson seismographs and possibly for each seismotectonic region.

The results of the present work are summarized in the diagram of figure (2) showing the variation of the examined magnitude scales with M_W . It is important to notice that no magnitude other than M_W is capable to express the "size" of earthquakes with $M_W > 8.0$ and that the available data do not permit for weak earthquakes ($M_W < 3.5$) the implicit estimation of moment magnitudes from magnitude measurements in other scales.

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